EPISTEMIC AGENCY UNDER DETERMINISM

Bridging Locus of Control, Epistemic Scope, and Empirical Decision Auditing

Theoretical & Empirical Working Paper

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ABSTRACT

This paper presents a unified theoretical and empirical framework reconciling causal determinism with practical human agency. We introduce the *Scope of System* model to distinguish between third-person omniscient determinism (global scope) and first-person bounded computation (agent scope). Building on compatibilist philosophy and locus of control theory, we demonstrate that acting under an internal locus of control remains the computationally optimal operational state for a decision-making agent, regardless of whether that mental state was pre-determined by antecedents. Finally, we operationalize this synthesis through the *Daily Decision-Tracking Protocol (DDTP)*—a micro-longitudinal methodology designed to measure decision latency, evaluate bounded rationality, and cultivate metacognitive precision.

I. INTRODUCTION AND LITERATURE REVIEW

The tension between causal determinism and personal agency represents one of the most enduring paradoxes in philosophy and cognitive science. Causal determinism posits that all mental states, choices, and neurological outputs are strictly dependent functions of prior physical states, genetic architecture, and historical environmental inputs (Hume, 1748). A prevalent psychological fallacy stemming from this proposition is fatalism: the assumption that if future events are pre-determined by antecedent conditions, individual cognitive effort is superfluous, rendering subjective agency meaningless.

However, conflating causal determinism with psychological fatalism represents a fundamental category mistake. Psychological literature demonstrates that an individual's cognitive orientation toward control—specifically their *Locus of Control*—profoundly dictates behavioral efficiency and objective outcome quality (Rotter, 1966). Agents possessing an internal locus of control exhibit enhanced problem-solving capacity, emotional resilience, and executive function compared to those operating under an external locus.

Compatibilist frameworks (Frankfurt, 1971) and pragmatic epistemologies (James, 1897) suggest that personal agency operates validly within local computational boundaries. Rather than viewing determinism as an impediment to action, this paper synthesizes compatibilist metaphysics, informational epistemic scope, and behavioral auditing into a unified framework. We establish that first-person agency is a required operational step within a deterministic causal chain and introduce an empirical diagnostic tool to audit real-world rational decision-making.

II. THEORETICAL MODEL 1: THE EPISTEMIC "SCOPE OF SYSTEM"

To resolve the contradiction between pre-determined physical trajectories and the necessity of real-time choice, we introduce the formal distinction between two informational boundaries: the *Omniscient Global Scope* (S_{global}) and the *Agent Local Scope* (S_{local}).

2.1 The Omniscient Scope (S_{global})

Let S_{global} represent an idealized system boundary defined by complete informational state access (analogous to Laplace's Demon). For any initial state vector at time t, total state information I_t strictly yields all future trajectory states F_{t+n} through physical transition function f:

$$F_{t+n} = f(I_t)$$

Within S_{global}, counterfactual possibilities do not exist; the state space trajectory is invariant. However, S_{global} is physically and computationally inaccessible to localized biological architectures due to thermodynamic entropy and physical capacity constraints.

2.2 The Agent Scope (S_{local})

In contrast, S_{local} defines the operational computational boundary of the individual decision-making subject. Because an agent cannot possess antecedent global state data nor compute its own neural trajectory prior to real-time execution, the future remains uncalculated within S_{local}. The internal cognitive engine must construct probabilistic models over potential future states (F_A, F_B, ..., F_N):

$$P(F_k \mid S_{local}) > 0 \quad \text{for all candidate choices } k$$

The physical process of choosing constitutes the exact computational routine required to collapse internal candidate probabilities F_k into a singular realized state. From the scope of S_{local}, decision-making is not a superfluous illusion but the mandatory physical mechanism through which future states are computed and actualized.

III. THEORETICAL MODEL 2: DETERMINISTIC AGENCY AND LOCUS OF CONTROL

We next examine the internal mechanics of psychological control within a deterministic framework. An agent's subjective locus of control acts as a critical parameter within its decision algorithm, directly modulating output efficiency.

LOCUS STATE	COMPUTATIONAL ASSUMPTION	BEHAVIORAL & ALGORITHMIC OUTPUT
Internal Locus (L_{int})	Internal processing and subjective effort are essential causal inputs for target outcomes.	High cognitive engagement, proactive optimization, superior task efficiency and resilience.
External Locus (L_{ext})	External forces decouple internal effort from final state outcomes.	Learned helplessness, degraded deliberation, suboptimal execution, systemic cognitive passivity.

Even if an agent's adoption of L_{ext} or L_{int} is strictly determined by past antecedents, operating under L_{int} yields superior algorithmic performance. Using determinism to justify L_{ext} represents a category error: it imports an uncalculable external frame (S_{global}) into an active first-person computational unit (S_{local}), artificially degrading output efficiency.

IV. EMPIRICAL FRAMEWORK: THE DAILY DECISION-TRACKING PROTOCOL (DDTP)

To bridge these theoretical models with empirical behavioral science, we propose the *Daily Decision-Tracking Protocol (DDTP)*. This methodology operationalizes Ecological Momentary Assessment (Shiffman et al., 2008) and dual-process cognitive theories (Kahneman, 2011) to audit decision validity and eliminate cognitive fallacies.

4.1 Three-Phase Protocol Execution

1. **Phase 1: Morning Baseline Mapping (Cold Cognition):** Prior to environmental exposure, the subject logs known daily parameters, explicit targets, and initial situational models.
2. **Phase 2: Real-Time Choice & Latency Logging:** Real-world choices are tracked in natural environments along with *Decision Latency* (T_d). Measuring T_d allows researchers to distinguish fast, heuristic System 1 processing from deliberate System 2 analysis.
3. **Phase 3: Evening Retrospective Audit:** The subject reviews deviations between morning intent and real-time execution. Deviations are categorized via debriefing as either *Rational Adaptations* (valid pivots responding to novel environmental data) or *Cognitive Fallacies* (fatigue, present bias, emotional rationalization).

V. DISCUSSION AND STRATEGIC IMPLICATIONS

The DDTP provides an empirical feedback mechanism that shifts agents from fatalistic external loci to high-efficiency internal loci. By demonstrating that internal computation directly alters real-world state trajectories within S_{local}, subjects achieve metacognitive alignment without violating deterministic physical laws.

References

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